



Create a cute picture for my HCI class where I am teaching user stories. [ChatGPT]

CSE472: HCI – Human Computer Interaction

Lecture – 8: User Stories

Spring 2026



Inspiring Excellence

User Stories

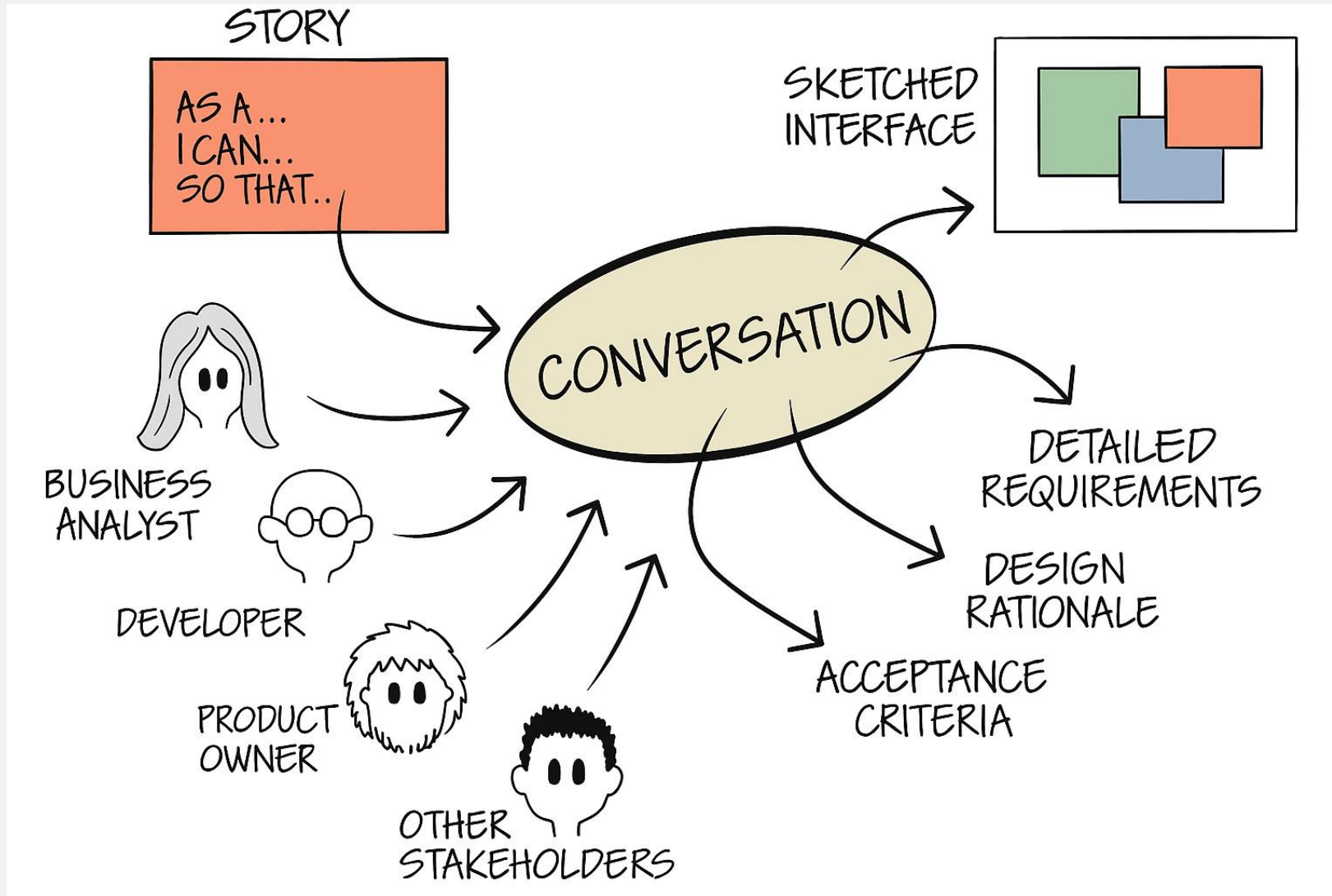
- Last time: We learned to create personas
- We will now use these personas to define user stories.
- User stories are short, simple descriptions of a software feature written from the end-user's perspective to ensure the product meets their needs.
- They follow the format "As a [user], I want [an action] so that [I can achieve a goal]" and are used to guide the design, development, and testing process by focusing on user value rather than technical details.

User Stories

User stories:

- Describe functionality that will be valuable to user of software.
- A feature request from a user's point of view.
- It is NOT a bug report or a functional specification.
- It is NOT a technical design document.

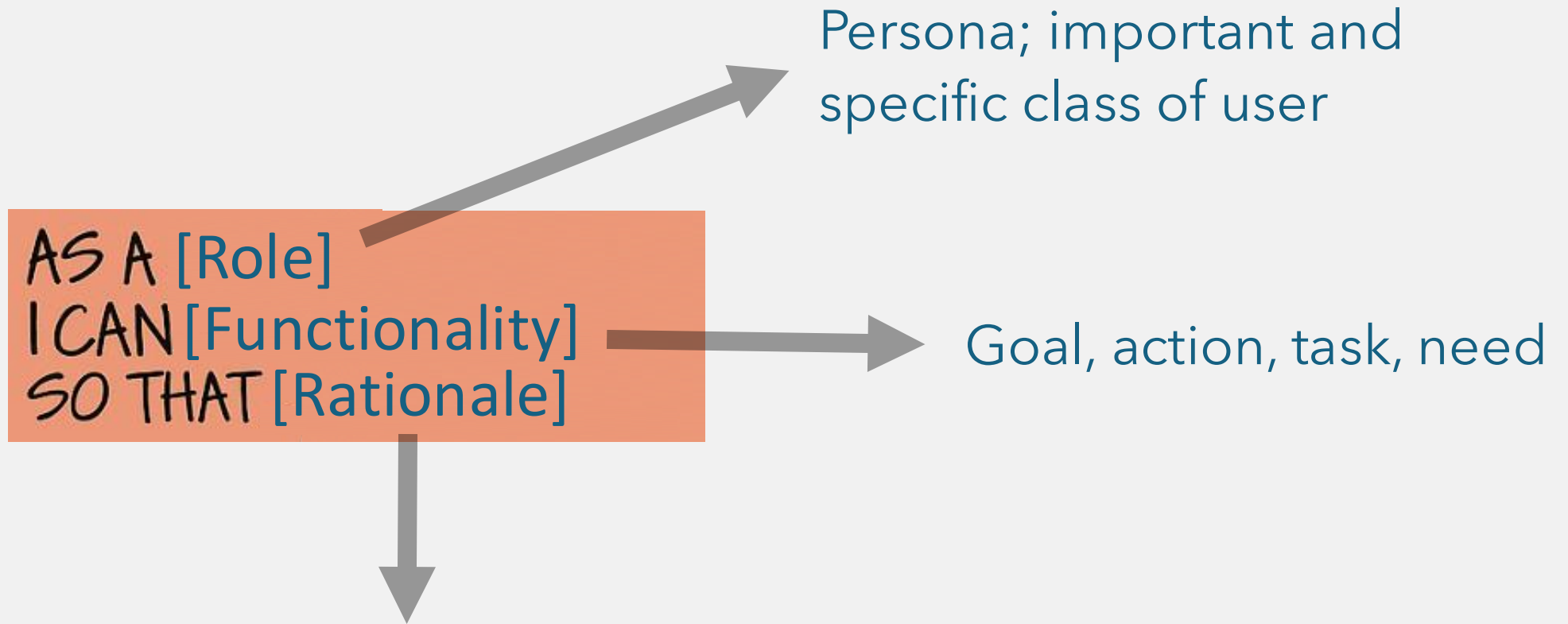
User Stories as a Design Tool



User Stories as a Design Tool

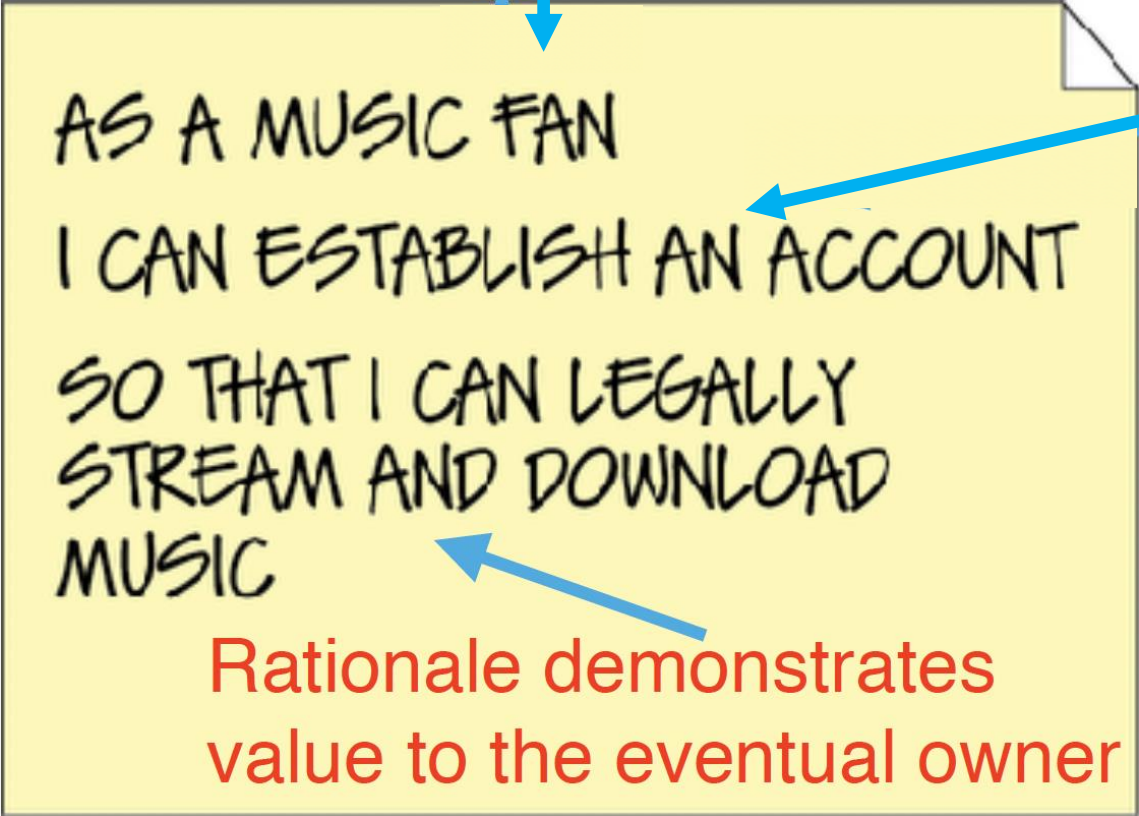
- Shifts the focus from the product to what the eventual owner / user needs or wants it to do.
- A customer buys your product to help them do something.
- Your stories must help them do it.
- A user has to do something using your product.
- Your stories must help them do it.

Component of a User Story



Example

Role is important to
the eventual owner



AS A MUSIC FAN
I CAN ESTABLISH AN ACCOUNT
SO THAT I CAN LEGALLY
STREAM AND DOWNLOAD
MUSIC

Functionality
enables the value
to be achieved

Rationale demonstrates
value to the eventual owner

Story Scope

- Too broad = impossible to test/code
- Too narrow = more time spent specifying than implementing
- Split long stories ("epics") into smaller pieces
- Rather than specify small details, get those in conversations with customer & annotate story
- Big stories can serve as placeholders for areas of the system that still need to be discussed

Story Scope

"As an administrator I can manage posts to the site so that I can control the content our customers see"

- manage?
- what content?
- control?



This is called
an Epic

Split into
smaller stories

Writing Stories

- Customer writes stories
 - Written in language of business to allow prioritization
 - Customer is primary product visionary
- Good user stories follow the INVEST principle to make them easy to plan, build, and test.
 - **I**ndependent
 - **N**egotiable
 - **V**aluable to users or customers
 - **E**stimable
 - **S**mall
 - **T**estable

Independent

- Each story has its own value
- Don't make stories depend on other stories
 - Stories that depend on other stories are difficult to prioritize and estimate

Example (Food Delivery App)

Bad (dependent stories)

- Story A: As a user, I want to create an account.
- Story B: As a user, I want to order food after logging in.
- Story B **depends on Story A.**
- If Story A is delayed, Story B also stops.

Better approach

- Break them so each still has value.

Example:

- As a user, I want to **browse restaurant menus** so that I can see available food.
- As a user, I want to **create an account to place an order.**
- Now one does not completely block the other.

Negotiable

- Stories serve as reminders not contracts
- Details come from conversation
- A user story is usually one sentence. But developers, designers, and product owners discuss the details later. So the story can change during discussion.

Example:

User Story:

As a user, *I want* to search restaurants *so that* I can find food quickly.

Questions to discuss:

Search by restaurant name?

Search by food type?

Search by location?

These details are negotiated during discussion.

Valuable

- Should be valuable for both to people using the software and paying for the software
- Avoid stories valued only by developers (make the benefits to customers/users apparent for these stories)

Example:

Bad Approach:

- Developer-focused story
"All connections to the database must use connection pooling."
- Students or customers **don't understand the value.**

Good Approach:

Rewrite to show user value

- *As a* user, *I want* the system to support 50 users simultaneously *so that* the app does not slow down during busy hours.
- Now the benefit is clear.
- *As a* customer, *I want* to track my delivery in real time *so that* I know when my food will arrive.
- Clear value for users.

Estimable

- We must have enough information that we can properly size a story so that we plan properly and commit to our work.
- Reasons why story might not be estimable:
 - Designers lack domain knowledge
 - Get details from customer
 - Story is too big
 - Split the story into smaller ones

Example:

Bad Approach:

- As a user, I want a better app experience.
- Too vague.
- Developers **cannot estimate time.**

Good Approach:

- As a user, I want to filter restaurants by rating.
- As a user, I want to sort restaurants by delivery time.
- Now developers can estimate: 1 day, 2 days etc.

Small

- User Stories should be small enough that they are able to be completed within a sprint. (e.g., A few hours to a few days of design/development time.)
- Makes them easy to use in planning
- Split compound & complex stories
- Combine too small stories

Testable

- All members of the team need a clear and precise way to verify whether or not a User Story has been completed.
- Can't tell if story is done without tests
- Tests can be automatable or not

Example

User Story:

- As a user, I want to [search restaurants by name](#).

Test cases:

- User types restaurant name
- Matching restaurants appear
- If none found → show "No results"
- Now the team can [verify if the feature works](#).

Examples

AS A BANK CUSTOMER
I CAN CHECK MY
ACCOUNT ONLINE
SO THAT I CAN SEE MY
BALANCE 24/7

This technological
solution hides
the real
requirement

Why does anyone want to
know their current balance?

A Better Story

AS A BANK CUSTOMER
I CAN SEE MY AVAILABLE
BALANCE AND ALL
ANTICIPATED PAYMENTS
FOR THE MONTH SO THAT I
CAN KNOW HOW MUCH IS
AVAILABLE TO ME BEFORE
NEXT PAYDAY

Guidelines for Good Stories

- Start with goal stories
- Write closed stories (stories that have a definite end point)
 - “A recruiter can review resumes from applicants to one of her ads” instead of “A recruiter can manage the ads she has placed”
- Size your story appropriately for the time frame it may be implemented in
- Don't rely solely on stories if some things are better expressed in other ways
- Include user roles/personas in stories rather than saying “user”
- Write for a single user (“A Job Seeker” not “Job Seekers”)
- Use Active Voice

Prioritizing Stories

- Assign points to stories based on difficulty/ time required
- For each release, customer prioritizes stories
- Developers determine velocity (# of points per release cycle) for previous cycle and plan to implement the highest priority stories up to that number of points for the release

Story responsibilities: Designer

- Help the customers write stories that
 - Are promises to converse rather than detailed specs
 - Have value to the users or the customer
 - Are independent
 - Are testable
 - Are appropriately sized
- Describing the need for technology/infrastructure in terms of value to users or customers
- Have the conversations with the customers

Story responsibilities: Customer

- Writing stories that
 - Are promises to converse rather than detailed specs
 - Have value to users or to yourself
 - Are independent
 - Are testable
 - Are appropriately sized
- Have the conversations with the designers/developers

Let's Fix These Examples

AS A STUDENT I WANT TO CHECK MY GRADES ONLINE
SO THAT I DON'T HAVE TO WAIT UNTIL THE MORNING

AS A STUDENT I CAN LEARN MY GRADES IMMEDIATELY
THEY ARE AVAILABLE SO THAT I DON'T SUFFER LONGER
THAN NECESSARY

Let's Fix These Examples

AS A STUDENT I WANT THE WEBSITE TO LOOK NICE

As a student

I want a clear dashboard showing my courses and grades
so that I can easily navigate the portal.

Let's Fix These Examples

AS A PRODUCT OWNER I WANT TO DRAG AND DROP ALL THE PRODUCT BACKLOG ITEMS SO THAT I CAN EASILY ADJUST PRIORITIES

AS A PRODUCT OWNER I CAN BE MADE AWARE OF THE HIGHEST PRIORITY NEEDS SO THAT I CAN ALIGN BACKLOG ITEMS WITH PRIORITIES

Summary: Why User Stories?

- Emphasize verbal communication
 - Comprehensible by everyone
 - Right size for planning
 - Good for iterative development
 - Encourage deferring detail
 - Encourage participatory design
 - Build up tacit knowledge
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- Your story must be right for the solution to be right.
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- Think about the non-functional needs – usability, convenience, security, cultural and so on.

Reading

- From Personas to User Stories

<https://www.romanpichler.com/blog/personas-epics-user-stories/>

- 10 tips for writing good user stories

<https://www.romanpichler.com/blog/10-tips-writing-good-user-stories/>

- User Stories

<https://www.interaction-design.org/literature/topics/user-stories>

Any questions?